Basins of Attraction of Periodic Orbits of Rational Maps on the Riemann Sphere: Algorithms, Bounds, and Visualizations

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Abstract

We study the basins of attraction of periodic attracting and superattracting orbits of rational self-maps $f\colon \mathbf{P}^1(\mathbb{C}) \to \mathbf{P}^1(\mathbb{C})$ of degree $d \geq 2$, using homogeneous coordinates throughout so that the point at infinity is treated on equal footing with finite points. A central tool is the *spherical derivative* $f^\#$, whose geometric mean around a periodic orbit gives the *Lyapunov constant* c; a cycle is attracting when $c < 1$ and superattracting when $c = 0$.

Our main contributions are threefold. First, we provide a self-contained algorithmic framework for locating and classifying all attracting p-cycles (periodic orbits of minimal period p) and their basins in $\mathbf{P}^1(\mathbb{C})$, based on the homogeneous model and the chordal metric. Second, we prove and exploit Fatou's critical-orbit theorem: every attracting cycle attracts at least one of the $2d-2$ critical points of f, yielding a sharp and effectively computable upper bound P on the maximal attracting period. Third, we implement these algorithms in the Julia package `RiemannBasins.jl` and illustrate them on several families of rational maps, including the quadratic family $z^2 + c$, Newton's method for polynomials, and Halley's method.

The connection between the Wronskian of the homogeneous pair (F, G) and the correct two-chart formula for $f^\#$ is discussed in detail; a subtle sign issue with the standard single-chart formula near ∞ is identified and corrected.

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1 Introduction

The dynamics of rational maps on the Riemann sphere is one of the central topics of one-dimensional complex analysis. The Fatou set $F(f)$ of a rational map f of degree $d \geq 2$ is the maximal open subset of $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ on which the family of iterates $\{f^n\}_{n \geq 0}$ is locally equicontinuous; its complement $J(f)$ is the Julia set. By Sullivan's No-Wandering-Domain theorem [19], every connected component of $F(f)$ is eventually

periodic, and the periodic Fatou components fall into exactly five types: attracting basins, superattracting basins, parabolic basins, Siegel discs, and Herman rings [15].

Among these, the *attracting and superattracting basins* are both the most common in practice and the most amenable to algorithmic computation, because the orbit of any point in such a basin converges geometrically (or super-geometrically) to the attracting cycle. Given a p-cycle $\Gamma = \{z_0, z_1 = f(z_0), \ldots, z_{p-1} = f^{p-1}(z_0)\}$ with $f(z_{p-1}) = z_0$, the *basin of attraction* $B_f(\Gamma) = \{z \in \mathbf{P}^1(\mathbb{C}) : f^{np}(z) \to \Gamma\}$ decomposes into p pairwise disjoint f-invariant open sets B_{z_k}, one for each cycle point z_k; each B_{z_k} may itself have infinitely many connected components (the cycle point, all its preimages under f^p, preimages of preimages, etc.). The *immediate basin* $B^0_{z_k}$ is the connected component of B_{z_k} that contains z_k itself. The map f sends $B^0_{z_k}$ onto $B^0_{z_{k+1 \bmod p}}$, so the p immediate basin components form a single cycle of simply connected domains.

The key practical question is: *how do we find all attracting cycles and their basins, and how many iterations suffice?* Fatou's theorem [11, 12] provides the answer: the immediate basin $B^0_{z_0} \cup \cdots \cup B^0_{z_{p-1}}$ of every attracting cycle contains at least one *critical point* of f (a zero of the Wronskian $\mathrm{Wr}(F, G) = F'G - FG'$ in homogeneous coordinates). Since f has at most $2d - 2$ critical points, it can have at most $2d - 2$ distinct attracting or superattracting cycles, and tracing the orbit of each critical point reveals every such cycle.

Our implementation in the Julia package `RiemannBasins.jl` [3] exposes this theory through two related function groups. The first, `basin_portrait_pcycle`, visualises the basin of every p-cycle for a user-supplied period p. The second, the `critical_orbit_analysis` family (§14 of the package), implements Fatou's theorem directly: it computes the Wronskian roots, traces each critical orbit, detects the period of the attracting cycle reached, computes the Lyapunov constant c, and returns a sharp upper bound P via `max_attracting_period`. The Lyapunov-based cycle-detection strategy originates in the theoretical framework of [2] and was first prototyped in `LCD.jl` [1]; a further application to the dynamical study of Chebyshev's method on polynomials appears in [4].

The topological structure of the Julia set $J(f)$ provides an independent perspective on the fractal geometry underlying the basin boundaries studied here. For an expanding rational map, the shape of $J(f)$—in the sense of Mardešić–Segal shape theory—is captured by an inverse system of cubic complexes built from the global (spherical) multiplier; the resulting shape invariants and the d-fold self-overlay structure of the restriction $f|_{J(f)} \colon J(f) \to J(f)$, together with its symmetries encoded via the fundamental pro-groupoid in the symmetric group Σ_d, are studied in [9, 8].

Several Julia packages for complex dynamics visualisation already exist, most notably `Fatou.jl` [18], which accepts arbitrary complex maps (including transcendental ones) and produces richly coloured escape-time and orbit-trap portraits with GPU acceleration. `RiemannBasins.jl` differs in three respects. First, it works on the full Riemann sphere $\mathbf{P}^1(\mathbb{C})$ in homogeneous coordinates, so the point $\infty = [1 : 0]$ is a regular point with its own basin and its own affine chart, rather than a mere escape threshold. Second, it implements Fatou's critical-orbit theorem (Theorem 4.4) algorithmically: given the homogeneous pair $[F : G]$, the package locates *all* attracting cycles automatically via Wronskian root-finding, computes their Lyapunov constants, and returns a provably sharp upper bound P on the maximum attracting period. Third, its `pcycle_lch` colour strategy encodes the cyclic structure of the basin decomposition—which cycle, which sub-basin within that cycle, and how fast—using a perceptually uniform LCH_{ab} scheme, making the cyclic permutation $f \colon B_j \to B_{j+1 \bmod p}$ directly readable from the image (Section 6).

The paper is organised as follows. Section 2 reviews the homogeneous model for

$\mathbf{P}^1(\mathbb{C})$, the chordal metric, and the spherical derivative in both affine charts. Section 3 defines p-cycles, their Lyapunov stability, and the classification into attracting/superattracting/indeterminate orbits. Section 4 summarises the relevant theory of Fatou components and states Fatou's critical-orbit theorem. Section 5 presents the two main algorithms in full pseudocode. Section 7 illustrates the algorithms on four families of maps. Section 8 discusses open problems and extensions.

2 Preliminaries: the Riemann sphere in homogeneous coordinates

2.1 The projective line $\mathbf{P}^1(\mathbb{C})$

We model the Riemann sphere as the *complex projective line*

$$\mathbf{P}^1(\mathbb{C}) = \big(\mathbb{C}^2 \setminus \{(0,0)\}\big)/\!\sim, \qquad (z,t) \sim (\lambda z, \lambda t), \quad \lambda \in \mathbb{C}^*,$$

writing a point as a homogeneous pair $[z : t]$. The affine chart $\{t \neq 0\}$ is identified with $\mathbb{C}$ via $[z : t] \mapsto z/t$, and the point $[1 : 0]$ represents ∞. We work with *normalised* representatives: given $(z,t) \neq (0,0)$, define

$$\nu(z,t) = \begin{cases} (z/t,\ 1) & \text{if } |t| \geq |z|, \\ (1,\ t/z) & \text{if } |z| > |t|. \end{cases} \tag{1}$$

The first case gives coordinates in the standard affine chart $(\zeta, 1)$ with $|\zeta| \leq 1$; the second gives coordinates in the second chart $(1, w)$ with $|w| < 1$.

2.2 Chordal metric

The *chordal metric* on $\mathbf{P}^1(\mathbb{C})$ is

$$\sigma\big([z_1 : t_1], [z_2 : t_2]\big) = \frac{2\,|z_1 \bar{t}_2 - z_2 \bar{t}_1|}{\sqrt{(|z_1|^2 + |t_1|^2)(|z_2|^2 + |t_2|^2)}}. \tag{2}$$

It satisfies $0 \leq \sigma \leq 2$, with $\sigma = 2$ iff one point is $0 = [0 : 1]$ and the other is $\infty = [1 : 0]$. On normalised representatives $\nu(z,t) = (\zeta, 1)$, formula (2) reduces to the standard spherical distance $\sigma([\zeta_1 : 1], [\zeta_2 : 1]) = 2|\zeta_1 - \zeta_2|/\sqrt{(1 + |\zeta_1|^2)(1 + |\zeta_2|^2)}$.

2.3 Rational maps in homogeneous coordinates

A rational map $f \colon \mathbf{P}^1(\mathbb{C}) \to \mathbf{P}^1(\mathbb{C})$ of degree $d \geq 1$ is given by a *homogeneous pair* $[F : G]$ of polynomials

$$F(z,t) = \sum_{k=0}^{d} a_k\, z^k\, t^{d-k}, \quad a_k \in \mathbb{C}, \tag{3}$$

$$G(z,t) = \sum_{k=0}^{d} b_k\, z^k\, t^{d-k}, \quad b_k \in \mathbb{C}, \tag{4}$$

where F and G have no common homogeneous factor, and $f([z:t]) = [F(z,t) : G(z,t)]$. We adopt the *constant-first* convention: the coefficient list is $[a_0, a_1, \ldots, a_d]$ so that a_0 is the constant term (coefficient of t^d) and a_d is the leading term (coefficient of z^d). Under the affine identification $[z:1] \leftrightarrow z$, the map becomes $f(z) = p(z)/q(z)$ where $p(z) = F(z,1) = \sum_k a_k z^k$ and $q(z) = G(z,1) = \sum_k b_k z^k$.

Remark 2.1. When $\deg p \neq \deg q$, one polynomial is padded with leading zero coefficients until both have degree $d = \max(\deg p, \deg q)$. This ensures both F and G are homogeneous of the same degree, and in particular that $f(\infty) = [a_d : b_d]$ is well defined.

2.4 Spherical derivative — two-chart formula

The *spherical derivative* of f at $[z:t] \in \mathbf{P}^1(\mathbb{C})$ measures the local expansion in the chordal metric. In the affine chart ($|t| \geq |z|$, $u = z/t$) it equals

$$f^{\#}(u) = \frac{|f'(u)|\,(1+|u|^2)}{1+|f(u)|^2}, \tag{5}$$

while in the second chart ($|z| > |t|$, $w = t/z$) it equals $f^{\#} = g^{\#}(w)$ for $g(w) = 1/f(1/w)$.

In homogeneous coordinates, let

$$W_z(z,t) = \frac{\partial F}{\partial z}\,G - F\,\frac{\partial G}{\partial z}, \qquad\qquad W_t(z,t) = \frac{\partial F}{\partial t}\,G - F\,\frac{\partial G}{\partial t}.$$

Both are homogeneous of degree $2d-1$. Euler's identity for homogeneous functions gives $z\,\partial F/\partial z + t\,\partial F/\partial t = d\,F$, from which one derives

$$W_t(z,t) = -\frac{z}{t}\,W_z(z,t), \qquad t \neq 0. \tag{6}$$

Proposition 2.2 (Two-chart spherical derivative)**.** *For any normalised representative $\nu(z,t)$ of a point $[z:t] \in \mathbf{P}^1(\mathbb{C})$,*

$$f^{\#}([z:t]) = \begin{cases} \dfrac{|W_z(z,t)|}{|t|\,(|F(z,t)|^2 + |G(z,t)|^2)}\,(|z|^2+|t|^2) & \text{if } |t| \geq |z|, \\[2ex] \dfrac{|W_t(z,t)|}{|z|\,(|F(z,t)|^2 + |G(z,t)|^2)}\,(|z|^2+|t|^2) & \text{if } |z| > |t|. \end{cases} \tag{7}$$

For normalised representatives ($\max(|z|,|t|) = 1$) the denominators $|t|$ and $|z|$ both equal 1, so the formula simplifies to $f^{\#} = |W_{\text{chart}}|\,(|z|^2+|t|^2)/(|F|^2+|G|^2)$.

Proof. In the first chart, $t = 1$, $u = z$ and $W_z(u,1) = p'(u)q(u) - p(u)q'(u)$ is the affine Wronskian. By homogeneity of degree $2d-1$, $W_z(z,t) = t^{2d-1}W_z(z/t,1)$. Substituting into (5) and simplifying yields the first case. The second case follows by applying the first case to $g(w)$ in the second chart. □

Remark 2.3. Using W_z in both charts (as in a naive single-formula implementation) gives the correct value for first-chart points but returns 0 for points of the form $[1:0]$ (the point ∞) due to (6): $W_z(1,0) = 0$ whenever the leading coefficients of F and G conspire to make W_z vanish there, while $W_t(1,0)$ correctly captures the derivative in the second chart. An implementation must therefore branch on the chart; see Section 5 and the note in `RiemannBasins.jl` §14.

3 Periodic orbits and their stability

3.1 p-Cycles

Definition 3.1. A point $z_0 \in \mathbf{P}^1(\mathbb{C})$ is *periodic* of *(minimal) period* $p \geq 1$ under f if $f^p(z_0) = z_0$ and $f^k(z_0) \neq z_0$ for $1 \leq k < p$. The set $\Gamma_p = \{z_0, z_1, \ldots, z_{p-1}\}$ with $z_k = f^k(z_0)$ is called a *p-cycle*. A periodic point of period 1 is a *fixed point*.

Equivalently, the points $z_0, \ldots, z_{p-1}$ are the fixed points of f^p that are *not* fixed points of f^k for any proper divisor $k \mid p$ with $k < p$.

Remark 3.2 (Counting p-cycles)**.** The number of fixed points of f^p in $\mathbf{P}^1(\mathbb{C})$ is $d^p + 1$ (counted with multiplicity, by Bézout's theorem applied to the homogeneous system $F_p(z,t) = \lambda\, z$, $G_p(z,t) = \lambda\, t$). By Möbius inversion, the number of periodic points of exact minimal period p is $\sum_{k|p} \mu(p/k)(d^k + 1)$, where μ is the Möbius function.

3.2 Stability and the Lyapunov constant

The classical stability criterion for a p-cycle Γ_p uses the *multiplier*

$$\lambda(\Gamma_p) = (f^p)'(z_0) = \prod_{k=0}^{p-1} f'(z_k), \tag{8}$$

computed in the affine chart (or via the Schwarzian derivative near ∞). The cycle is *attracting* if $|\lambda| < 1$, *superattracting* if $\lambda = 0$, *repelling* if $|\lambda| > 1$, and *indifferent* if $|\lambda| = 1$.

For numerical and algorithmic purposes, the *Lyapunov constant* [2]

$$c(\Gamma_p) = \left(\prod_{k=0}^{p-1} f^{\#}(z_k) \right)^{1/p} \tag{9}$$

is more convenient, as it uses the chart-independent spherical derivative $f^{\#}$ and is defined uniformly for all cycle points including ∞.

Proposition 3.3. *Let $\Gamma_p = \{z_0, \ldots, z_{p-1}\}$ be a p-cycle with all points in the affine chart $\{t \neq 0\}$. Then*

$$c(\Gamma_p)^p = |\lambda(\Gamma_p)| \cdot \frac{\prod_k (1 + |z_k|^2)}{\prod_k (1 + |f(z_k)|^2)}.$$

Since $f(z_k) = z_{k+1}$ (cyclically), the last ratio equals 1, and therefore $c(\Gamma_p) = |\lambda(\Gamma_p)|^{1/p}$.

The proposition shows that $c < 1 \Leftrightarrow |\lambda| < 1$, $c = 0 \Leftrightarrow \lambda = 0$, etc., so the Lyapunov constant carries the same stability information as the multiplier, but is well-defined at ∞ and throughout $\mathbf{P}^1(\mathbb{C})$.

Definition 3.4. Let $c = c(\Gamma_p)$ be the Lyapunov constant of a p-cycle.

- $c = 0$: *superattracting* cycle.
- $0 < c < 1$: *attracting* cycle.
- $c \geq 1$ (after numerical warmup): *indeterminate* (may be repelling, indifferent, or the warmup was insufficient).

The threshold $c < 1$ requires a numerical tolerance; in practice, we use $c < \varepsilon_{\text{attr}} = 1 - 10^{-6}$ for attracting and $c < 10^{-8}$ for superattracting.

3.3 Superattracting cycles and critical points

A cycle Γ_p is superattracting if and only if at least one cycle point z_k satisfies $f'(z_k) = 0$, i.e., z_k is a *critical point* of f. In this case the spherical derivative $f^{\#}(z_k) = 0$, so the product in (9) vanishes.

Definition 3.5. A point $z_0 = [z : t] \in \mathbf{P}^1(\mathbb{C})$ is a *critical point* of $f = [F : G]$ of degree d if

$$W_z(z,t) = \frac{\partial F}{\partial z}(z,t)\, G(z,t) - F(z,t)\, \frac{\partial G}{\partial z}(z,t) = 0.$$

(If additionally $|z| > |t|$, one uses W_t instead; see Proposition 2.2.)

The polynomial $W_z(z,1) = p'(z)q(z) - p(z)q'(z)$ has degree at most $2d-2$, so there are at most $2d-2$ finite critical points. The point $\infty = [1:0]$ is a critical point if and only if $\deg(W_z) < 2d-2$, equivalently if the leading coefficient of $W_z(z,1)$ vanishes, which happens when f has a critical point at ∞ in the second-chart sense.

4 Fatou components and basins of p-cycles

4.1 The Fatou and Julia decomposition

Definition 4.1. The *Fatou set* $F(f)$ of f is the set of points $z \in \mathbf{P}^1(\mathbb{C})$ having a neighbourhood U on which $\{f^n|_U\}$ is a normal family (in the sense of Montel). Its complement $J(f) = \mathbf{P}^1(\mathbb{C}) \setminus F(f)$ is the *Julia set*.

Both $F(f)$ and $J(f)$ are completely invariant: $f^{-1}(F(f)) = F(f)$ and $f^{-1}(J(f)) = J(f)$. For $d \geq 2$ the Julia set is a non-empty perfect set (compact with no isolated points) [15]. The topological *shape* of $J(f)$, studied via inverse systems of cubic complexes built from the spherical multiplier, is analysed in [9, 8].

Theorem 4.2 (Sullivan, 1985 [19])**.** *Every connected component U of $F(f)$ is eventually periodic: there exist $n_0, m \geq 0$ with $f^{n_0+m}(U) \subseteq f^{n_0}(U)$.*

Periodic Fatou components are classified into five types:

(i) **Attracting basin**: contains an attracting cycle point.

(ii) **Superattracting basin**: contains a superattracting cycle point.

(iii) **Parabolic basin**: boundary contains an indifferent cycle with $\lambda = e^{2\pi i p/q}$ a root of unity.

(iv) **Siegel disc**: biholomorphic to a disc; f acts as an irrational rotation.

(v) **Herman ring**: biholomorphic to an annulus; f acts as an irrational rotation.

4.2 Basins of p-cycles

Definition 4.3. Let $\Gamma_p = \{z_0, \dots, z_{p-1}\}$ be an attracting or superattracting p-cycle. The *basin of attraction* of Γ_p is

$$B_f(\Gamma_p) = \big\{ z \in \mathbf{P}^1(\mathbb{C}) : \sigma(f^{np}(z), \Gamma_p) \to 0 \text{ as } n \to \infty \big\}. \tag{10}$$

The *immediate basin* of z_k is the connected component $B^0_{z_k}$ of $B_f(\Gamma_p)$ containing z_k.

The full basin is the disjoint union $B_f(\Gamma_p) = \bigsqcup_{k=0}^{p-1} B_{z_k}$ where $B_{z_k} = \{z : \sigma(f^{np+k}(z), z_k) \to 0\}$; note that f maps B_{z_k} to $B_{z_{k+1}}$ (indices mod p). The spherical measure (probability) of these basins on S^2, and its dependence on the relaxation parameter, was studied quantitatively in [13].

Theorem 4.4 (Fatou, 1919–1920 [11, 12])**.** *Let f be a rational map of degree $d \geq 2$ and let Γ_p be an attracting or superattracting p-cycle with immediate basins $B^0_{z_0}, \ldots, B^0_{z_{p-1}}$. At least one immediate basin component $B^0_{z_k}$ contains a critical point of f; equivalently, at least one critical point of f is attracted to Γ_p. (For superattracting cycles, every z_k is itself a critical point of f, so each $B^0_{z_k}$ contains one; for merely attracting cycles, at least one component suffices.)*

Proof sketch. Fix one immediate basin component $B^0_{z_0}$ and consider the iterate $g = f^p \colon B^0_{z_0} \to B^0_{z_0}$, which has an attracting fixed point at z_0. By the Riemann mapping theorem, $B^0_{z_0}$ is conformally equivalent to the unit disc $\mathbb{D}$ (it is a simply connected proper subset of $\mathbf{P}^1(\mathbb{C})$ when $d \geq 2$). Suppose for contradiction that g has no critical point in $B^0_{z_0}$. Then g is an unramified cover of its image, hence—since $B^0_{z_0} \cong \mathbb{D}$ is simply connected—a conformal automorphism of $\mathbb{D}$. Every conformal automorphism of $\mathbb{D}$ with an interior fixed point is a rotation, but a rotation has $|g'(z_0)| = 1$, contradicting $|g'(z_0)| = |(f^p)'(z_0)| = |\lambda|^p < 1$. Therefore g has at least one critical point $w \in B^0_{z_0}$. A critical point of $g = f^p$ at w means $(f^p)'(w) = 0$, so $f'(f^k(w)) = 0$ for some $k \in \{0, \ldots, p-1\}$: the point $f^k(w) \in B^0_{z_k}$ is a critical point of f. □

Corollary 4.5. *Since f has at most $2d-2$ critical points (counted with multiplicity), it has at most $2d-2$ distinct attracting or superattracting cycles. In particular, the maximum attracting period satisfies $P \leq 2d-2$.*

Corollary 4.6 (Completeness of critical orbit analysis)**.** *Tracing the orbit of every critical point of f under iteration and recording the period of any attracting cycle reached is* complete*: the maximum over all critical orbits equals the true maximum attracting period P.*

4.3 Böttcher coordinates near superattracting cycles

Near a superattracting fixed point z_0 (i.e., $f(z_0) = z_0$, $f'(z_0) = 0$) of local degree $m \geq 2$, the *Böttcher coordinate* ϕ linearises f as a degree-m map:

$$\phi(f(z)) = \phi(z)^m, \qquad \phi(z_0) = 0. \tag{11}$$

The Böttcher coordinate exists and is univalent on the immediate basin $B^0_{z_0}$ if and only if $B^0_{z_0}$ is simply connected, which holds when $B^0_{z_0}$ contains exactly one critical value [15].

For a superattracting p-cycle, one applies the Böttcher theory to f^p at each cycle point.

4.4 Kœnigs linearization near attracting cycles

Böttcher coordinates apply when the multiplier $\lambda = 0$ (superattracting case). For a merely attracting fixed point z_0 with $0 < |\lambda| < 1$, the analogous result is due to Kœnigs (1884).

Theorem 4.7 (Kœnigs, 1884)**.** *Let f be a holomorphic map with an attracting fixed point z_0 and multiplier $\lambda = f'(z_0)$, $0 < |\lambda| < 1$. Then there exists a unique conformal map $\phi \colon B^0_{z_0} \to \mathbb{D}$ with $\phi(z_0) = 0$, $\phi'(z_0) = 1$, satisfying the Schröder equation*

$$\phi(f(z)) = \lambda\, \phi(z), \qquad z \in B^0_{z_0}. \tag{12}$$

The map ϕ is obtained as the uniform limit $\phi(z) = \lim_{n\to\infty}(f^n(z) - z_0)/\lambda^n$.

For a p-cycle $\{z_0, \ldots, z_{p-1}\}$ one applies Theorem 4.7 to f^p at each cycle point, with multiplier $\lambda^p = (f^p)'(z_0)$. The Schröder equation shows that orbits in $B^0_{z_0}$ converge to z_0 at exactly the geometric rate $|\lambda|^n$: in the ϕ-coordinate the dynamics is a linear contraction by λ.

Together, Theorems 4.7 and **??** (Böttcher) complete the linearization picture for attracting cycles:

$$\begin{array}{rlll} \lambda = 0 & \text{(superattracting):} & \phi \circ f = \phi^m & \text{(Böttcher, doubly-exponential convergence),} \\ 0 < |\lambda| < 1 & \text{(attracting):} & \phi \circ f = \lambda\, \phi & \text{(Kœnigs, geometric convergence).} \end{array}$$

The iteration count $n(z)$ used in the lightness formula (14) is the number of steps to bring $|\phi(z)|$ below a fixed threshold, so it directly measures the ϕ-coordinate distance to the attractor.

5 Algorithmic framework

We now describe the two main algorithms implemented in `RiemannBasins.jl`: critical orbit analysis and basin portrait for p-cycles.

5.1 Homogeneous iteration and normalisation

A key primitive is the evaluation of $f^n([z : t])$ using normalised representatives. Given the homogeneous pair $[F : G]$, one step of iteration is

$$(z, t) \;\longmapsto\; \nu\big(F(z,t), G(z,t)\big), \tag{13}$$

where ν is the normalisation (1). Normalising at each step prevents overflow/underflow for orbits approaching ∞ or 0.

5.2 Algorithm 1: Critical orbit analysis

Algorithm 1 implements Corollary 4.6: it locates all critical points, traces their orbits, detects the attracting period (if any), and computes the Lyapunov constant.

Definition 5.1 (Maximum attracting period)**.** Given f of degree d, the *maximum attracting period bound* is

$$P = \max\big\{p(\phi_j) : \phi_j \text{ critical},\ \text{label}(\phi_j) \in \{\texttt{:attracting}, \texttt{:superattracting}\}\big\},$$

with $P = 0$ if no critical orbit reaches an attracting cycle. By Corollary 4.5, no attracting cycle of f has period exceeding P.

5.3 Algorithm 2: Basin portrait for p-cycles

Given a fixed period p, Algorithm 2 computes a colour-coded portrait of the basins of all attracting p-cycles.

Remark 5.2 (Two-chart portrait)**.** To visualise the full Riemann sphere without distortion near ∞, the portrait is drawn in both the standard chart $[z : 1]$ and the second chart $[1 : w]$, side by side, as implemented by `two_chart_portrait` in `RiemannBasins.jl`.

Algorithm 1 Critical orbit analysis for $f = [F : G]$ of degree d

Require: Coefficient lists `num`, `den` of F, G; parameters N_{iter}, $P_{\max}$, `tol`.
Ensure: List of `CriticalOrbitResult` structs.

1: **Step 1: Critical points.** Compute the affine Wronskian $W(z) = p'(z)q(z) - p(z)q'(z)$ as a polynomial. Find its roots $\{w_1, \ldots, w_m\}$ numerically (e.g. via the companion matrix). Optionally check whether $\infty = [1 : 0]$ is also critical ($\deg W < 2d - 2$). Convert each root to a normalised homogeneous pair $\nu(w_j, 1)$. Deduplicate via chordal distance $< 10^{-\texttt{prec}}$.

2: **Step 2: Warmup.** For each critical point ϕ_j:

   a. Set $\phi_j^{(0)} = \phi_j$.

   b. Iterate $\phi_j^{(n+1)} = \nu(F(\phi_j^{(n)}), G(\phi_j^{(n)}))$ for $n = 0, 1, \ldots, N_{\text{warm}} - 1$ where $N_{\text{warm}} = \lfloor N_{\text{iter}}/2 \rfloor$.

   c. If $|\phi_j^{(N_{\text{warm}})}| < \varepsilon_{\text{sing}}$, abort (super-zero hit).

3: **Step 3: Period detection.** Set $y = \phi_j^{(N_{\text{warm}})}$. For $q = 1, 2, \ldots, P_{\max}$:

   a. Compute $y_q = f^q(y)$ by applying (13) q times.

   b. If $\sigma(y, y_q) < 10^{-5}$, record period $p \leftarrow q$ and break.

   If no period found, set $p \leftarrow 0$ (indeterminate).

4: **Step 4: Collect cycle points.** Starting from y, collect $\{y, f(y), f^2(y), \ldots, f^{p-1}(y)\}$.

5: **Step 5: Lyapunov constant.** Compute

$$c = \exp\left(\frac{1}{p} \sum_{k=0}^{p-1} \log \max(f^{\#}(y_k),\, 10^{-300})\right),$$

   using the two-chart formula (7).

6: **Step 6: Label.**

- $c < 10^{-8}$: label $\leftarrow$ `:superattracting`.
- $c < 1 - 10^{-6}$: label $\leftarrow$ `:attracting`.
- Otherwise: label $\leftarrow$ `:indeterminate`.

7: **return** list of structs $(\phi_j, \text{label}, p, \{y_0, \ldots, y_{p-1}\}, c)$.

Remark 5.3 (Complexity)**.** The bottleneck of Algorithm 2 is the classification of M^2 grid points, each requiring $O(N)$ evaluations of f. With $N = O(d \cdot \log(1/\varepsilon))$ iterations (geometric convergence at rate $|\lambda| < 1$), the total cost is $O(M^2 d \log(1/\varepsilon))$ per chart. In the superattracting case the convergence is faster (doubly exponential for an m-th order superattracting cycle: $|\phi(f^n(z))| = |\phi(z)|^{m^n}$), so fewer iterations suffice.

6 Colour encoding for p-cycle basin portraits

A basin portrait for a rational map f of degree d must convey at least three independent pieces of information at each pixel simultaneously:

1. *Which attracting cycle* does the orbit of the pixel converge to?

2. *Which element of that cycle* is the orbit closest to after the transient?

Algorithm 2 Basin portrait for p-cycles of $f = [F : G]$

Require: Homogeneous pair $[F : G]$; period p; grid resolution $M \times M$; iteration budget N; colour palette.

Ensure: Colour image: basin assignment for each grid point.

1: **Step 1: Compute f^p.** Build the coefficient lists of the homogeneous pair $[F_p : G_p]$ representing f^p, by composing $[F : G]$ with itself p times symbolically (or evaluate numerically for each grid point).

2: **Step 2: Locate p-cycles.** Find all fixed points of f^p in $\mathbf{P}^1(\mathbb{C})$ by solving $F_p(z, 1)/G_p(z, 1) = z$ (affine chart) plus check at ∞. Among these, retain only those of exact minimal period p. Group into p-cycles $\{\Gamma_p^{(1)}, \Gamma_p^{(2)}, \ldots\}$. Retain only attracting/superattracting cycles (check $|\lambda|$ or c).

3: **Step 3: Assign colours.** Assign a distinct colour κ_j to each cycle $\Gamma_p^{(j)}$. The background (Julia set / indeterminate) colour is black.

4: **Step 4: Classify grid points.** For each grid point z_0 (expressed as $[z_0 : 1]$ in the first affine chart, supplemented by a second-chart grid for the global portrait):

   a. Iterate $z_0, f(z_0), f^2(z_0), \ldots$ up to N steps.

   b. At each iterate, check proximity to each p-cycle point using chordal distance $< \varepsilon$.

   c. If proximity is detected to cycle $\Gamma_p^{(j)}$, assign colour κ_j and move to next grid point.

   d. If N steps are exhausted without convergence, assign background colour.

5: **return** colour array.

3. *How fast* does the orbit converge (i.e. how many iterations are needed to reach a prescribed tolerance)?

The first two items identify the *sub-basin*: if f has a p-periodic attracting cycle $\Gamma_p = \{a_0, a_1, \ldots, a_{p-1}\}$ with $f(a_j) = a_{j+1 \bmod p}$, then its full basin $\mathcal{B}(\Gamma_p)$ splits into p dynamically equivalent components

$$\mathcal{B}_j(\Gamma_p) = \big\{z \in \hat{\mathbb{C}} : \operatorname{dist}(f^{np+r}(z),\, a_j) \to 0 \text{ for some } r\big\}, \quad j = 0, \ldots, p-1,$$

cyclically permuted by f: $f(\mathcal{B}_j) \subset \mathcal{B}_{j+1 \bmod p}$. A colour scheme that ignores this decomposition loses the internal structure of the basin and fails to make the cyclic symmetry of f visible.

6.1 The LCH_{ab} perceptual colour space

Standard RGB or HSV colour spaces are not perceptually uniform: equal numerical steps along a coordinate do not produce equally perceptible colour differences. We work instead in LCH_{ab}, the polar form of CIELAB [10], with coordinates

$$(L,\, C,\, H): \quad L \in [0, 100] \text{ (lightness)}, \quad C \geq 0 \text{ (chroma)}, \quad H \in [0^\circ, 360^\circ) \text{ (hue angle)}.$$

Perceptual uniformity means that the Euclidean distance ΔE_{ab} in CIELAB is a reliable proxy for perceived colour difference, so equal steps in L, C or H produce equally visible changes. Julia's `Colors.jl` library provides `LCHab` natively and handles the conversion to RGB for display [14].

6.2 Three-level encoding scheme

We assign colour to a pixel according to the following hierarchy.

Level 1 (which cycle) — hue H. Each attracting cycle $\Gamma_p^{(k)}$ is assigned a *base hue* $H^{(k)}$ drawn from the Okabe–Ito palette [16] converted to LCH_{ab}. The Okabe–Ito hues are well separated in perceptual space and are robust under common types of colour-vision deficiency. For up to seven cycles the base hues are approximately $H^{(0)} = 29°$ (orange), $H^{(1)} = 194°$ (sky blue), $H^{(2)} = 56°$ (yellow), $H^{(3)} = 162°$ (bluish green), $H^{(4)} = 300°$ (reddish purple), $H^{(5)} = 14°$ (vermillion), $H^{(6)} = 220°$ (blue).

Level 2 (which sub-basin) — hue shift ΔH. The j-th sub-basin of cycle k receives hue

$$H_j^{(k)} = H^{(k)} + j\,\frac{\Delta}{p}, \qquad j = 0, \ldots, p-1,$$

where $\Delta = 25°$ is a fixed angular bandwidth. This keeps all p sub-basins within the same perceptual hue family (the observer reads "these belong to the same cycle") while making them mutually distinguishable: in LCH_{ab} with $C \approx 55$, hue differences below 15° are barely perceptible, so a step of $25°/p$ is resolvable for $p \le 5$ and can be extended to $\Delta = 35°$ for larger periods. The cyclic order of the hue shifts mirrors the dynamics: f maps $\mathcal{B}_j$ to $\mathcal{B}_{j+1 \bmod p}$, shifting the hue by Δ/p in a fixed direction.

Level 3 (convergence speed) — lightness L. Let $n(z)$ be the number of iterations of f required for the orbit of z to enter a prescribed tolerance ball around Γ_p. We set

$$L(z) = L_{\max} - (L_{\max} - L_{\min})\left(\frac{n(z)}{n_{\max}}\right)^{\gamma}, \qquad \gamma = \tfrac{1}{2}, \tag{14}$$

with $L_{\max} = 85$ and $L_{\min} = 28$. The square-root exponent $\gamma = \frac{1}{2}$ applies a perceptual gamma correction: it compresses the lightness scale near the attractor (where n is small and many pixels cluster) and expands it near the Julia set (where n is large and fine structure appears). The chroma $C = 55$ is held constant across all pixels.

Coherence with $p = 1$ (fixed points). When $p = 1$ there is a single sub-basin ($j = 0$), no hue shift occurs, and the scheme reduces exactly to the `positionplusiteration` strategy already implemented: base colour from the Okabe–Ito cycle identifier, luminance from iteration count. The p-cycle encoding is thus a strict extension of the existing convention.

6.3 Implementation in `RiemannBasins.jl`

The scheme is implemented as the colour strategy `"pcycle_lch"` in `basin_portrait_pcycle`. Once the orbit of a pixel has been classified as converging to cycle k, the sub-basin index j is determined by identifying which element of Γ_p is closest to the final iterate:

$$j(z) = \arg\min_{0 \le j < p} \sigma\big(f^N(z),\, a_j\big)\,,$$

where σ is the chordal metric and N is the total iteration count. The colour is then

$$\mathrm{colour}(z) = \mathrm{RGB}\Big(\mathrm{LCH}_{ab}\big(L(z),\ 55,\ H_{j(z)}^{(k)}\big)\Big)\,. \tag{15}$$

Remark 6.1. The choice $\Delta = 25°$ is a default; the user may override it via the keyword argument `hue_band` in `basin_portrait_pcycle`. For families with many simultaneous cycles one may also reduce Δ and increase C slightly to maintain separability.

7 Examples and visualizations

We illustrate the algorithms on six families of rational maps, with particular attention to the quadratic family $f_c(z) = z^2 + c$. All computations use `RiemannBasins.jl` [3].

7.1 The quadratic family $f_c(z) = z^2 + c$ and the Mandelbrot set

The simplest non-trivial one-parameter family of degree-2 maps is $f_c(z) = z^2 + c$, $c \in \mathbb{C}$. Its unique finite critical point is $z_0 = 0$ (Wronskian $W(z) = 2z$), so by Fatou's theorem (Theorem 4.4) the entire structure of attracting dynamics is controlled by the single orbit $\{0,\, c,\, c^2 + c,\, \ldots\}$.

The *Mandelbrot set* is defined as

$$\mathcal{M} = \bigl\{ c \in \mathbb{C} : \{f_c^n(0)\}_{n \geq 0} \text{ is bounded} \bigr\}. \tag{16}$$

A parameter c lies in $\mathcal{M}$ if and only if the critical orbit does not escape to ∞; equivalently, $|f_c^n(0)| \leq 2$ for all $n \geq 0$. The exterior of $\mathcal{M}$ is coloured according to the *normalised escape time*

$$\nu(c) = n - \log_2 \max\bigl(1,\, \log_2 |f_c^n(0)|\bigr), \tag{17}$$

where n is the first iterate at which $|f_c^n(0)| > 2$; this smooth colouring removes banding artefacts [17].

The interior of $\mathcal{M}$ decomposes into *hyperbolic components* H_p: maximal connected open sets on which f_c has an attracting cycle of period exactly p. By Fatou's theorem, $c \in H_p$ if and only if the orbit of the critical point 0 converges to an attracting p-cycle. The largest component H_1 is the main cardioid (attracting fixed point); the large circular bulb H_2 attached to its left is the "head" (attracting 2-cycle); H_1 also carries a countable family of bulbs at each rational internal angle r/s, whose period is determined by the Farey structure of $\mathcal{M}$ [6].

The *centre* of H_p is the unique parameter $c_* \in H_p$ satisfying $f_{c_*}^p(0) = 0$: the attracting cycle of f_{c_*} is *superattracting*, and Algorithm 1 returns `:superattracting` with period p. All four quadratic examples in this section correspond to such centres (Figure 1, Table 1).

Name	Parameter c_*	Period p	Location in $\mathcal{M}$
Basílica	-1	2	head bulb H_2
Airplane	≈ -1.7549	3	real-axis bulb of H_1
Douady rabbit	$\approx -0.1226 + 0.7449i$	3	upper bulb of H_1
Period-5 centre	$\approx -0.5043 + 0.5628i$	5	angle-1/5 bulb of H_1

Table 1: The four centres of hyperbolic components of $\mathcal{M}$ studied in this section. Each is a superattracting parameter ($f_{c_*}^p(0) = 0$); their locations are marked in Figure 1.

Theorem 7.1 (Douady–Hubbard, 1982 [6])**.** *For the quadratic family $f_c(z) = z^2 + c$, the following are equivalent:*

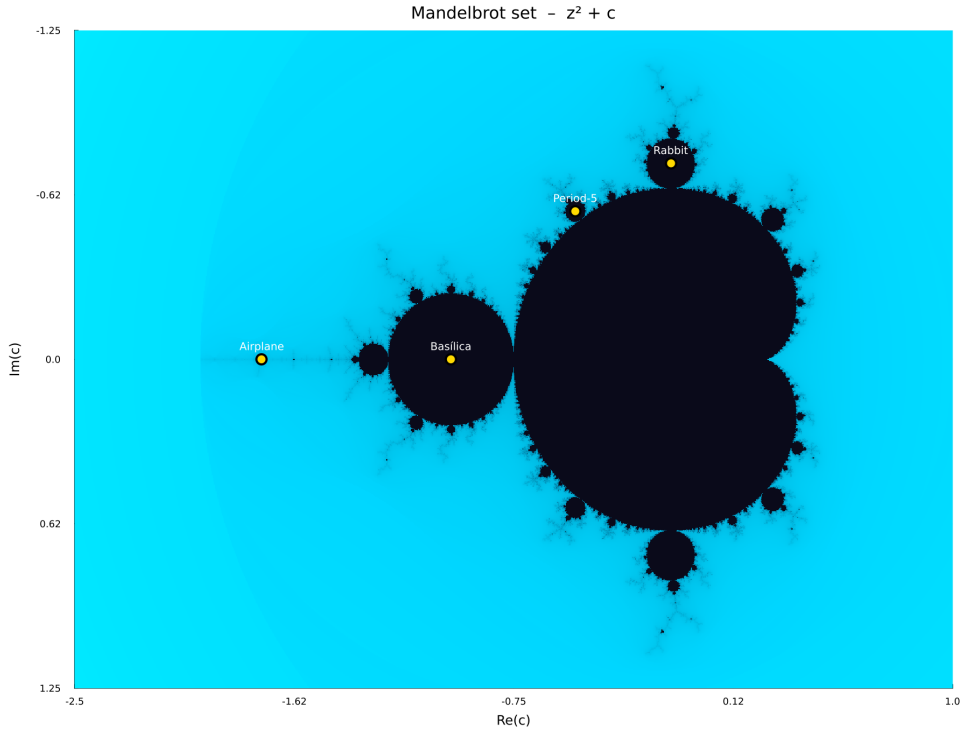


Figure 1: The Mandelbrot set $\mathcal{M}$ rendered by `mandelbrot_portrait` in `RiemannBasins.jl`. The interior is shown in near-black; the exterior is shaded by normalised escape time (17) (blue gradient). The four centres of Table 1 are marked with yellow dots: Basílica ($c = -1$), Airplane ($c \approx -1.7549$), Douady rabbit ($c \approx -0.1226 + 0.7449i$), and Period-5 centre ($c \approx -0.5043 + 0.5628i$). Every marked point is a superattracting parameter; the corresponding Julia-set basin portraits are shown in Sections 7.2–7.8.

(i) *The Julia set J_c is connected.*

(ii) *The filled Julia set $K_c = \{z \in \mathbb{C} : \{f_c^n(z)\}_{n\geq 0}$ is bounded$\}$ is connected.*

(iii) *The critical orbit $\{0, c, c^2+c, \ldots\}$ is bounded, i.e. $c \in \mathcal{M}$.*

If $c \notin \mathcal{M}$, then J_c is a Cantor set and every immediate basin is simply connected (in fact, $K_c = J_c$).

Proof idea. The key tool is the *Böttcher coordinate at ∞*: since ∞ is a superattracting fixed point of f_c, there exists a conformal map $\Phi_c \colon B_\infty^0 \to \{|w| > 1\}$ conjugating f_c to $w \mapsto w^2$. The map Φ_c extends to the boundary of B_∞^0 if and only if no critical value lies in B_∞^0, i.e. $0 \notin B_\infty^0$, i.e. $c \in \mathcal{M}$. When $c \in \mathcal{M}$, Φ_c extends to $\partial B_\infty^0 = J_c$, showing J_c is connected. When $c \notin \mathcal{M}$, $0 \in B_\infty^0$ and iterating the Böttcher coordinate shows J_c is a Cantor set. □

All four examples of this article ($c = -1$, $c \approx -1.7549$, $c \approx -0.1226 + 0.7449i$, $c \approx -0.5043 + 0.5628i$) lie inside $\mathcal{M}$, so their Julia sets are connected and each immediate basin $B_{z_k}^0$ is simply connected. This is precisely what makes the basin portraits of Figures 2–9 consist of simply connected domains with no "holes".

External rays and the Green's function. The *Böttcher coordinate at ∞* of f_c defines the *Green's function*

$$G_c(z) = \lim_{n\to\infty} \frac{1}{2^n} \log |f_c^n(z)|, \qquad z \in B_\infty^0,$$

which satisfies $G_c(f_c(z)) = 2G_c(z)$ and $G_c(z) \to 0$ as $z \to \partial B_\infty^0 = J_c$. Level sets $\{G_c = r\}$ are *equipotential curves*; gradient lines of G_c are the *external rays* $\mathcal{R}_c(\theta)$ (at angle $\theta \in [0, 1)$). When $c \in \mathcal{M}$, every rational ray $\mathcal{R}_c(p/q)$ lands at a point of J_c (Douady–Hubbard); the combinatorics of landing points encodes the hyperbolic component structure of $\mathcal{M}$ and connects the symbolic dynamics of angle-doubling to the topology of J_c. The normalised escape time (17) is precisely $G_c(c)/\log 2$ (up to additive normalisation), so the smooth colouring of the exterior in Figure 1 is an approximation of the Green's function, and the visible "bands" before smoothing are the equipotential curves.

7.2 The quadratic family $f_c(z) = z^2 + c$ (Basílica, $c = -1$)

The map $f_{-1}(z) = z^2 - 1$ has degree $d = 2$, so $2d - 2 = 2$ critical points. In homogeneous notation: $F(z,t) = z^2 - t^2$, $G(z,t) = t^2$. The affine Wronskian is $W(z) = p'q - pq' = 2z$, with unique root $z_0 = 0$. Since $\deg W = 1 < 2$, the point $\infty = [1 : 0]$ is also critical.

In the second chart $w = 1/z$, $g(w) = w^2/(1-w^2)$ and $g'(0) = 0$, so ∞ is a *superattracting* fixed point of f_{-1}.

The orbit of the finite critical point $z_0 = 0$:

$$0 \overset{f}{\longmapsto} -1 \overset{f}{\longmapsto} 0 \overset{f}{\longmapsto} -1 \longmapsto \cdots$$

This reveals the *basílica 2-cycle* $\Gamma_2 = \{0, -1\}$. Since $z_0 = 0 \in \Gamma_2$ and $f'(0) = 0$, the cycle is superattracting with $c = 0$. By Algorithm 1, $P = 2$.

The two basins B_0 and B_{-1} partition the Fatou set (minus the basin of ∞) into two equal components; their common boundary is the bas'ilica Julia set, a dendrite-like connected curve (see Figure 2).

Example 7.2 (Basílica computation).

```
using RiemannBasins
num = complex([-1.0, 0.0, 1.0]);  den = complex([1.0])
print_critical_orbit_summary(num, den)
# → P = 2 (superattracting 2-cycle, c = 0.0)
basin_portrait_pcycle(num, den, 2; iter=400, show_plot=true)
```

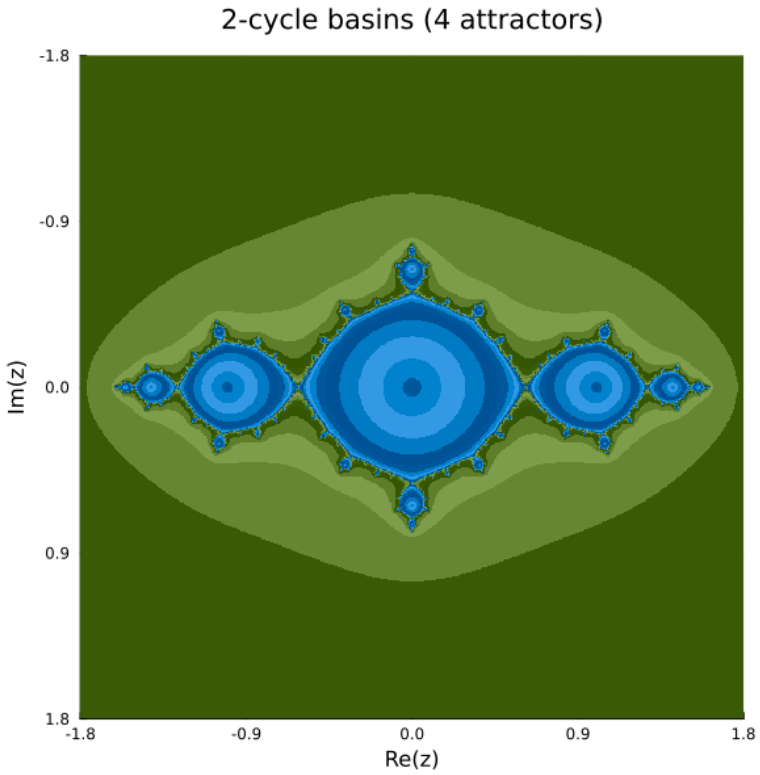


Figure 2: Basin portrait of the 2-cycle of $f_{-1}(z) = z^2 - 1$ (basílica). Each colour represents the basin of one element of the superattracting 2-cycle $\{0, -1\}$; the grey region is the basin of ∞.

Figure 2 follows the classical convention: each colour is assigned to one *orbit group*, so both sub-basins of the same cycle share a common hue family. The `pcycle_lch` strategy of Section 6 goes one step further and encodes which sub-basin each pixel belongs to via a hue shift ΔH. For $p = 2$ with `hue_band`$= 120°$ the two sub-basins $B_0 \ni 0$ and $B_{-1} \ni -1$ are separated by $60°$ within the same perceptual hue family, making the cyclic permutation $f\colon B_0 \to B_{-1} \to B_0$ directly readable from the image (Figure 3).

7.3 Newton's method for $z^3 - 1$

Newton's method $N(z) = \frac{2z^3+1}{3z^2}$ has degree $d = 3$. The Wronskian $W(z) = 6z^2 \cdot 3z^2 - (2z^3 + 1) \cdot 6z = 6z(z^3 - 1)$. The four critical points are $z = 0$ and $z \in \{1, \omega, \bar{\omega}\}$ where $\omega = e^{2\pi i/3}$.

The three roots of unity are superattracting fixed points of N (Newton converges quadratically). The critical point $z = 0$ maps to $N(0) = \infty$; in the second chart $g(w) = 3w/(2 + w^3)$, $g'(0) = 3/2 > 1$, so ∞ is *repelling* and the orbit of $z = 0$ is *indeterminate*.

The three superattracting fixed points give $P = 1$ (see Figure 4).

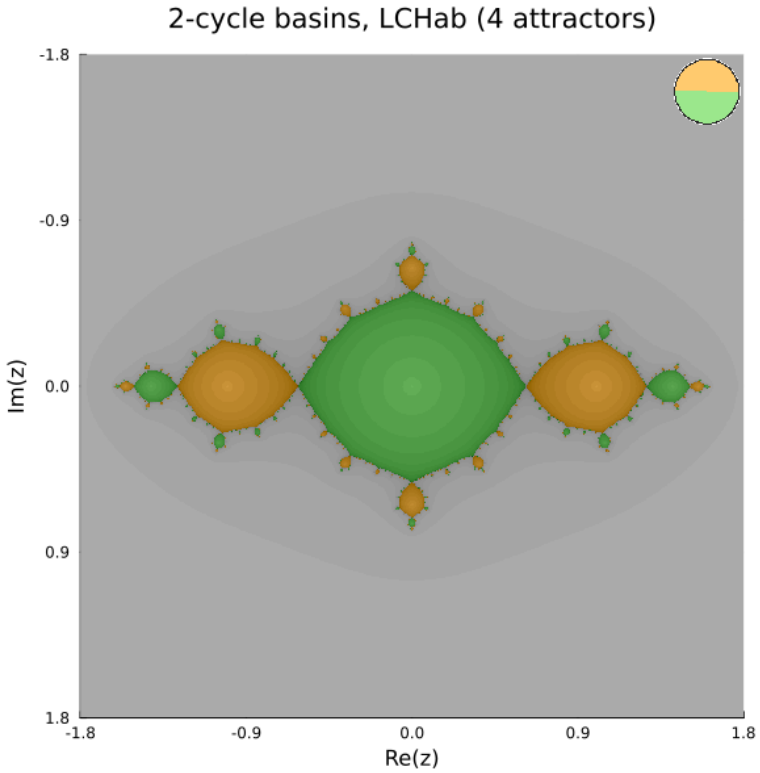


Figure 3: The same map $f_{-1}(z) = z^2 - 1$ rendered with the `pcycle_lch` strategy (`hue_band`= 120°). The two sub-basins B_0 (lighter hue) and B_{-1} (darker hue) are now distinguished by a 60° hue shift; their common boundary is the basílica Julia set.

Example 7.3 (Newton portrait).

```
num = complex([1.0, 0.0, 0.0, 2.0])
den = complex([0.0, 0.0, 3.0, 0.0])
print_critical_orbit_summary(num, den)
# → 3 superattracting fixed points (p=1), 1 indeterminate
# → P = 1
basin_portrait_pcycle(num, den, 1; iter=300, show_plot=true)
```

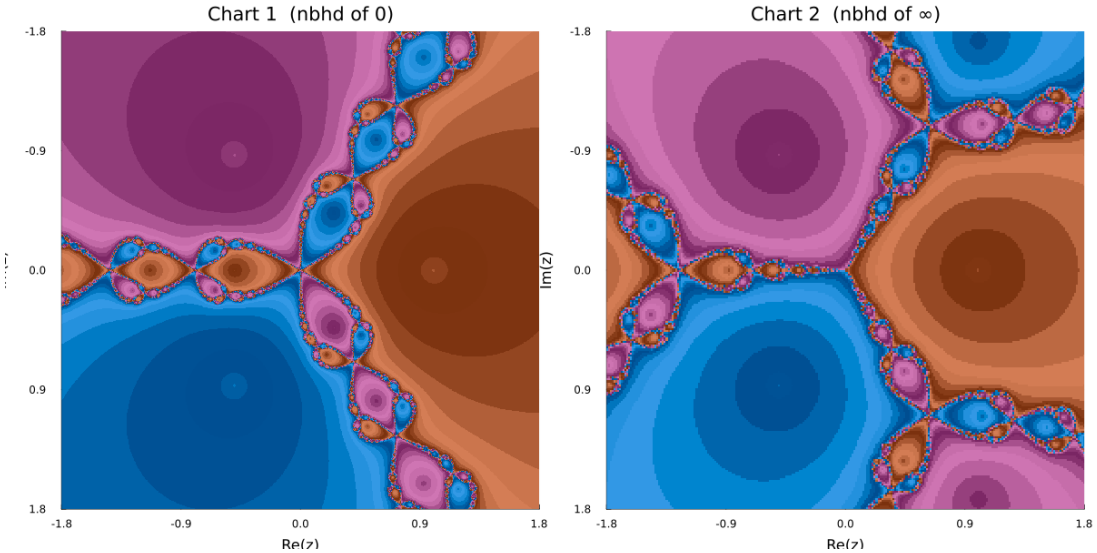


Figure 4: Basin portrait of Newton's method for $z^3 - 1$ in both affine charts of $\mathbf{P}^1(\mathbb{C})$. Chart 1 (left) shows the neighbourhood of the origin; Chart 2 (right) the neighbourhood of ∞. The three superattracting fixed points (roots of $z^3 - 1$) each attract a connected basin; ∞ is repelling.

7.4 Halley's method for $z^3 - 1$

Halley's method (1694) for solving $p(z) = 0$ is the second-order Householder iteration

$$H(z) = z - \frac{2p(z)p'(z)}{2[p'(z)]^2 - p(z)p''(z)}. \tag{18}$$

It achieves *cubic* convergence near a simple root (error cubed at each step), compared with quadratic for Newton.

Rational form for $p(z) = z^3 - 1$. With $p(z) = z^3 - 1$, $p'(z) = 3z^2$, $p''(z) = 6z$ formula (18) simplifies to

$$H(z) = \frac{z(z^3 + 2)}{2z^3 + 1} = \frac{z^4 + 2z}{2z^3 + 1}, \tag{19}$$

a rational map of degree $d = 4$. The Wronskian (in the affine chart) is

$$W(z) = P'Q - PQ' = (4z^3+2)(2z^3+1) - (z^4+2z)(6z^2) = 2(z^3-1)^2 = 2(z-1)^2(z-\omega)^2(z-\bar{\omega})^2,$$

where $\omega = e^{2\pi i/3}$. Thus H has $2d - 2 = 6$ critical points (counted with multiplicity), all coinciding with the three roots of unity $1, \omega, \bar{\omega}$, each of multiplicity 2.

Fixed-point analysis. The fixed points of H are found by solving $H(z) = z$:

$$z(z^3 + 2) = z(2z^3 + 1) \implies z^3 = 1 \quad (\text{plus } z = 0 \text{ and } z = \infty).$$

So H has five fixed points in $\mathbf{P}^1(\mathbb{C})$ $(= d + 1)$:

- $z = 1, \omega, \bar{\omega}$: superattracting (each is both a critical point and a fixed point, so the multiplier $H'(z_k) = 0$).
- $z = 0$: repelling, $H'(0) = 2(0 - 1)^2/(2 \cdot 0 + 1)^2 = 2$.
- $z = \infty$: repelling (in the second chart $\hat{H}(w) = w(2 + w^3)/(1 + 2w^3)$, $\hat{H}'(0) = 2$).

Since every critical point lies in an immediate basin of a superattracting fixed point, Fatou's theorem gives $P = 1$ and the dynamics is completely determined by the three basins.

The convergence near the roots is cubic rather than quadratic, which manifests in the basin portrait as smoother boundaries and a steeper lightness gradient away from each attractor. Compare Figure 5 with the Newton portrait (Figure 4): the basins are qualitatively similar but the Julia set of Halley is strictly contained in that of Newton's method (Halley converges for strictly more starting points), and the interior shading is more compressed near the attractors.

Example 7.4 (Halley portrait).

```
# H(z) = (z^4+2z)/(2z^3+1)
num = complex([0.0, 2.0, 0.0, 0.0, 1.0])   # z^4 + 2z
den = complex([1.0, 0.0, 0.0, 2.0])         # 2z^3 + 1
print_critical_orbit_summary(num, den)
# → 3 superattracting fixed points (p=1), 2 repelling fixed points
# → P = 1
basin_portrait_pcycle(num, den, 1; iter=300, colorstrategy="pcycle_lch",
                      show_plot=true)
```

7.5 The map $f(z) = z^2$ (superattracting fixed point)

For $f(z) = z^2$: $W(z) = 2z$, unique critical point $z = 0$, orbit $0 \to 0$. Since $f'(0) = 0$, $c = 0$ and $P = 1$. The basin of 0 is the open unit disc $\{|z| < 1\}$ and the Julia set is the unit circle $\{|z| = 1\}$ (see Figure 6).

7.6 The airplane map $f_c(z) = z^2 + c$ (3-cycle, $c \approx -1.7549$)

The map $f_c(z) = z^2 + c$ with c the unique real root of

$$c^3 + 2c^2 + c + 1 = 0, \qquad c \approx -1.7549,$$

has degree $d = 2$ and a single finite critical point $z_0 = 0$ (Wronskian $W(z) = 2z$). Evaluating:

$$f_c(0) = c, \quad f_c(c) = c^2 + c, \quad f_c(c^2 + c) = (c^2 + c)^2 + c = 0,$$

where the last equality holds precisely because c satisfies $c^3 + 2c^2 + c + 1 = 0$. Hence 0 lies on a *superattracting 3-cycle* $\Gamma_3 = \{0,\ c,\ c^2 + c\}$, and Algorithm 1 returns $P = 3$ (see Figure 7).

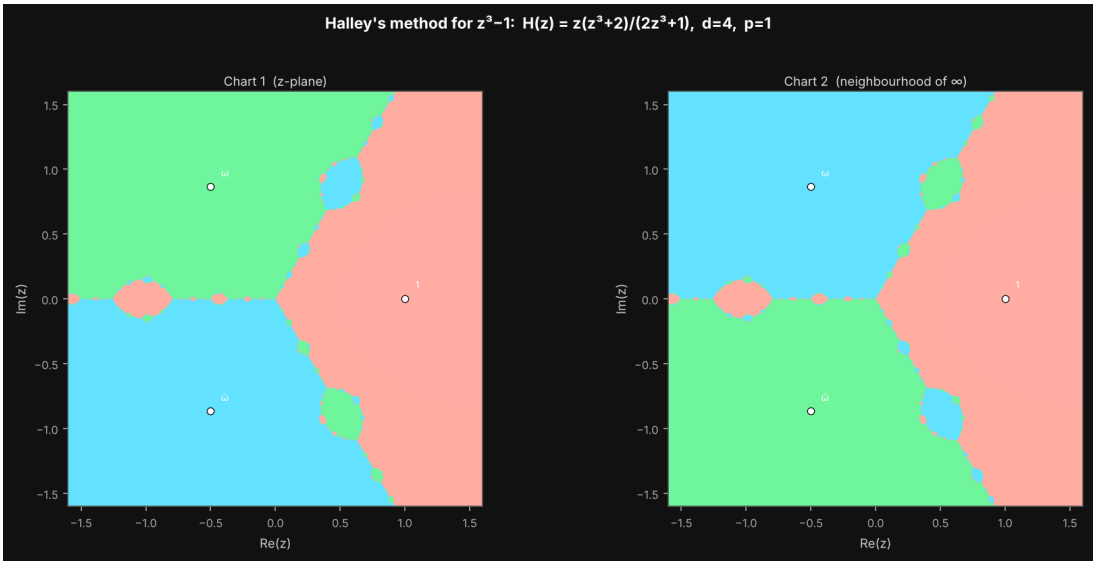


Figure 5: Basin portrait of Halley's method (19) for $z^3 - 1$, $d = 4$. Three superattracting fixed points $1, \omega, \bar{\omega}$ (roots of $z^3 - 1$) each of multiplicity 2; two repelling fixed points at $z = 0$ and $z = \infty$. Compare with Figure 4: Halley's basins cover a larger portion of the sphere.

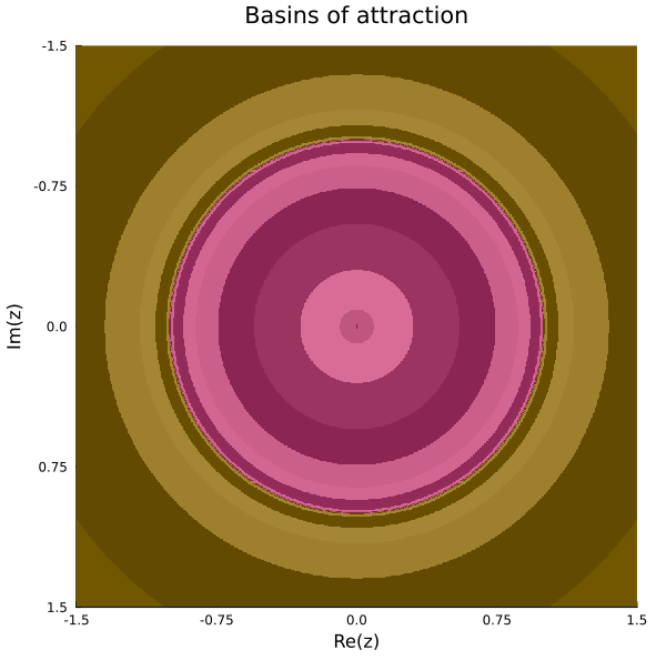


Figure 6: Basin portrait of $f(z) = z^2$. The open unit disc $\{|z| < 1\}$ (blue) is the basin of the superattracting fixed point $z = 0$; the exterior $\{|z| > 1\}$ is the basin of ∞. The unit circle $\{|z| = 1\}$ is the Julia set.

The corresponding Julia set—known as the *airplane*—is a connected, locally connected compact set with a characteristic "airplane wing" shape. The Fatou set decomposes into three basins B_0, B_c, B_{c^2+c}, one for each element of the 3-cycle, together with the basin of ∞.

Example 7.5 (Airplane computation).

```
using RiemannBasins
# c is the real root of c^3 + 2c^2 + c + 1 = 0
c = let x = -1.75
      for _ in 1:50; x -= (x^3+2x^2+x+1)/(3x^2+4x+1) end; x
    end
num = complex([c, 0.0, 1.0]);  den = complex([1.0])
print_critical_orbit_summary(num, den)
# → P = 3 (superattracting 3-cycle, c = 0.0)
basin_portrait_pcycle(num, den, 3; iter=400, show_plot=true)
```

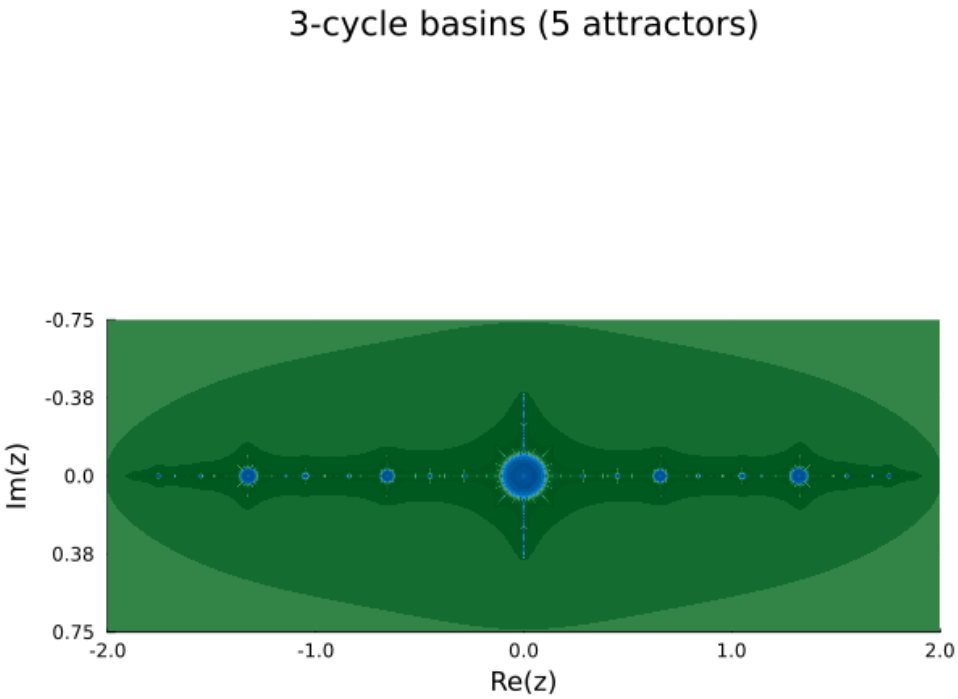


Figure 7: Basin portrait of the 3-cycle of the airplane map $f_c(z) = z^2 + c$ ($c \approx -1.7549$, the real root of $c^3 + 2c^2 + c + 1 = 0$). The three colours correspond to the basins of B_0, B_c and B_{c^2+c}; grey is the basin of ∞. The bounded components of the Fatou set form the characteristic "airplane" shape.

7.7 The Douady rabbit $f_c(z) = z^2 + c$ (3-cycle, $c \approx -0.1226 + 0.7449i$)

The same polynomial equation

$$c^3 + 2c^2 + c + 1 = 0$$

that determines the airplane parameter (Section 7.6) has two additional complex conjugate roots. Taking the root with positive imaginary part gives

$$c_{\mathrm{r}} \approx -0.12256116688 + 0.74486176662i,$$

the parameter of the *Douady rabbit* [6]. Since c_{r} satisfies the same equation, the critical orbit verification is identical:

$$f_{c_{\mathrm{r}}}(0) = c_{\mathrm{r}}, \quad f_{c_{\mathrm{r}}}(c_{\mathrm{r}}) = c_{\mathrm{r}}^2 + c_{\mathrm{r}}, \quad f_{c_{\mathrm{r}}}(c_{\mathrm{r}}^2 + c_{\mathrm{r}}) = 0,$$

so 0 lies on a superattracting 3-cycle $\Gamma_3 = \{0,\, c_{\mathrm{r}},\, c_{\mathrm{r}}^2 + c_{\mathrm{r}}\}$ and Algorithm 1 returns $P = 3$ as for the airplane.

Geometrically the two maps are strikingly different. The airplane Julia set has thin antenna-like tendrils; the Douady rabbit Julia set is a connected, locally connected compact set with three prominent *round petals*—the visual origin of its name [6]. These large, rounded sub-basins make the rabbit a cleaner illustration of the `pcycle_lch` colour encoding: with `hue_band`= 120° each of the three sub-basins B_0, $B_{c_{\mathrm{r}}}$, $B_{c_{\mathrm{r}}^2 + c_{\mathrm{r}}}$ receives a hue offset of 40° within the same perceptual hue family, producing three clearly distinguishable shades (see Figure 8).

Example 7.6 (Douady rabbit with LCH coloring).

```
using RiemannBasins
# c is the complex root of c^3 + 2c^2 + c + 1 = 0, Im(c) > 0
c_rabbit = let x = complex(-0.12, 0.75)
             for _ in 1:80
               x -= (x^3 + 2x^2 + x + 1) / (3x^2 + 4x + 1)
             end; x
           end
num = ComplexF64.([c_rabbit, 0.0, 1.0])
den = ComplexF64.([1.0])
basin_portrait_pcycle(num, den, 3;
                      iter = 400,
                      colorstrategy = "pcycle_lch",
                      hue_band = 120.0,
                      show_cyclic_palette = true,
                      xinterval = (-1.5, 1.5),
                      yinterval = (-1.5, 1.5))
```

7.8 Period-5 centre $f_c(z) = z^2 + c$ ($c \approx -0.5043 + 0.5628i$, 5-cycle)

The Mandelbrot set contains a hyperbolic component of period 5 attached to the main cardioid at internal angle 1/5. Its centre satisfies $f_c^{\circ 5}(0) = 0$ and is located at

$$c_5 \approx -0.50434017545 + 0.56276576145i.$$

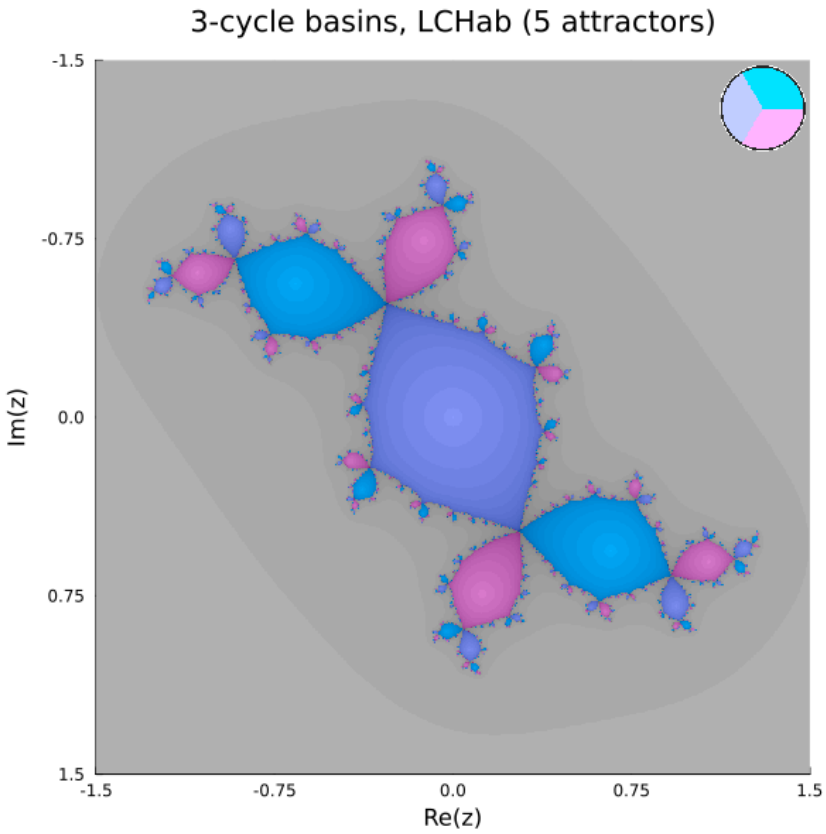


Figure 8: Basin portrait of the Douady rabbit $f_c(z) = z^2 + c$ ($c \approx -0.1226 + 0.7449i$, the complex root with positive imaginary part of $c^3 + 2c^2 + c + 1 = 0$) rendered with the `pcycle_lch` colour strategy (`hue_band`$= 120°$). The three sub-basins B_0, B_{c_r}, $B_{c_\mathrm{r}^2+c_\mathrm{r}}$ are separated by hue offsets of $40°$; grey is the basin of ∞. Compared with the airplane (Figure 7), the sub-basins are large and round, making the cyclic permutation $f\colon B_0 \to B_{c_\mathrm{r}} \to B_{c_\mathrm{r}^2+c_\mathrm{r}} \to B_0$ directly readable from the colours.

For this parameter the critical orbit closes after exactly five steps, so Algorithm 1 returns $P = 5$.

The five sub-basins $B_0, B_1, \ldots, B_4$ (indexed by the position of a pixel's limit point in Γ_5) are related by the cyclic permutation $f\colon B_k \to B_{k+1 \bmod 5}$. With `hue_band`$= 150^\circ$ the encoding assigns hue offsets $0^\circ, 30^\circ, 60^\circ, 90^\circ, 120^\circ$ to $B_0, \ldots, B_4$, yielding five perceptually well-separated shades within a single Okabe-Ito hue family (see Figure 9). The image reveals the five-fold rotational near-symmetry of the Julia set and makes the cyclic dynamics directly legible at a glance. The circular colour-wheel inset in the top-right corner displays the five sub-basin hues in cyclic order ($B_0 \to B_1 \to \cdots \to B_4 \to B_0$), serving as an on-figure legend for the colour encoding.

Example 7.7 (Period-5 centre).

```
using RiemannBasins
c5 = let x = complex(-0.5024, 0.5629)
       for _ in 1:100
         z = 0+0im; dz = 0+0im
         for _ in 1:5
           z, dz = z^2 + x, 2z*dz + 1
         end
         x -= z/dz
       end; x
     end   # = -0.50434...+0.56277...i
num = ComplexF64.([c5, 0.0, 1.0])
den = ComplexF64.([1.0])
basin_portrait_pcycle(num, den, 5;
                      iter = 500,
                      colorstrategy = "pcycle_lch",
                      hue_band = 150.0,
                      show_cyclic_palette = true,
                      xinterval = (-1.5, 1.6),
                      yinterval = (-1.2, 1.9))
```

7.9 A proper rational map: $f(z) = (z^2 - 1)/(z^2 + 1)$

All preceding examples are either polynomials or related to root-finding iteration. We now exhibit a *genuinely rational* function whose dynamics is not conjugate to any polynomial—here both 0 and ∞ play symmetric roles—and that has a superattracting 2-cycle.

The map and its critical points. Set $f(z) = (z^2 - 1)/(z^2 + 1)$, a rational map of degree $d = 2$. The Wronskian (affine chart) is

$$W(z) = P'Q - PQ' = 2z(z^2 + 1) - (z^2 - 1)2z = 4z,$$

so the only affine critical point is $z = 0$ (with multiplicity 1); the second critical point is $z = \infty$ (Wronskian degree $= 2d - 2 = 2$).

Superattracting 2-cycle. Direct calculation gives

$$f(0) = \frac{0 - 1}{0 + 1} = -1, \qquad f(-1) = \frac{1 - 1}{1 + 1} = 0,$$

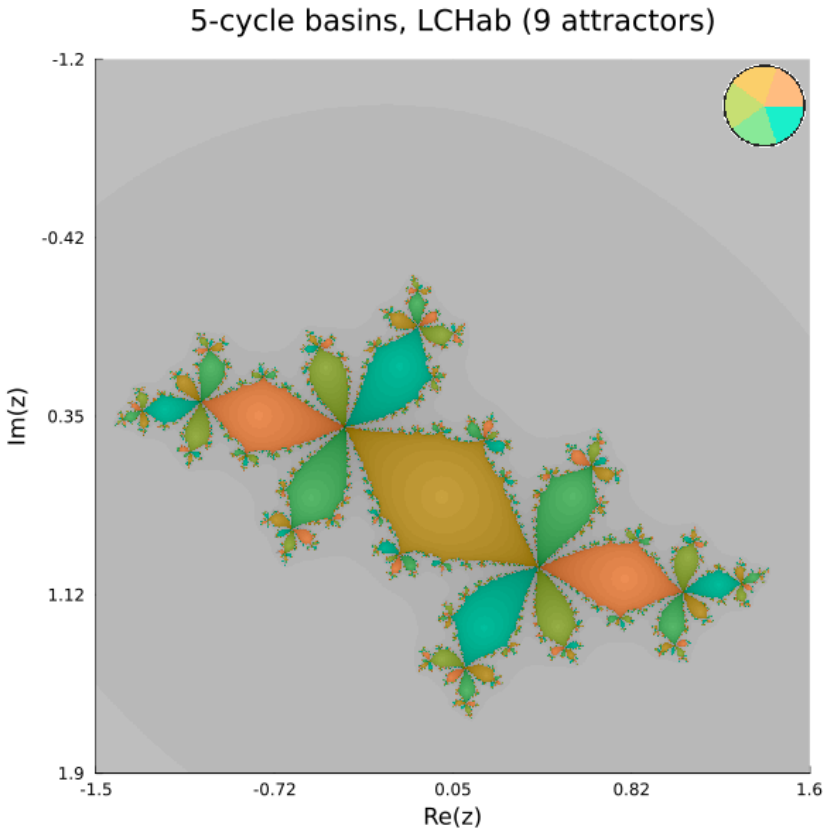


Figure 9: Basin portrait of the period-5 centre $f_c(z) = z^2 + c$ ($c \approx -0.5043 + 0.5628i$, superattracting 5-cycle) with the `pcycle_lch` colour strategy (`hue_band`$= 150°$, hue step $30°$ per sub-basin). The five sub-basins $B_0, \ldots, B_4$ are encoded by increasing hue within the same perceptual family; the five-fold near-symmetry of the Julia set and the cyclic permutation $f\colon B_0 \to B_1 \to \cdots \to B_4 \to B_0$ are readable directly from the colour progression. The circular colour-wheel inset (top right) shows the five sub-basin hues in their cyclic order.

so $\{0, -1\}$ is a 2-cycle. Since $f'(0) = 4 \cdot 0/(0+1)^2 = 0$, the cycle point $z_0 = 0$ is a critical point; hence $\{0, -1\}$ is a *superattracting* 2-cycle (Lyapunov constant $c(\{0, -1\}) = 0$).

The orbit of the second critical point: $f(\infty) = 1$, $f(1) = 0$, and from that point on the orbit follows the 2-cycle. Thus *both* critical points are attracted to $\{0, -1\}$, confirming by Corollary 4.5 that this is the *only* attracting cycle and $P = 2$.

Fixed points. Solving $f(z) = z$ gives $z^3 - z^2 + z + 1 = 0$, whose three solutions are all repelling (since both critical orbits converge to the 2-cycle, there is no critical point available to stabilize a fixed point).

Example 7.8 (Proper rational portrait).

```
# f(z) = (z^2 - 1) / (z^2 + 1)
num = complex([-1.0, 0.0, 1.0])   # z^2 - 1
den = complex([ 1.0, 0.0, 1.0])   # z^2 + 1
print_critical_orbit_summary(num, den)
# → 1 superattracting 2-cycle, 3 repelling fixed points
# → P = 2
basin_portrait_pcycle(num, den, 2; iter=400, colorstrategy="pcycle_lch",
                      hue_band=180.0, show_plot=true)
```

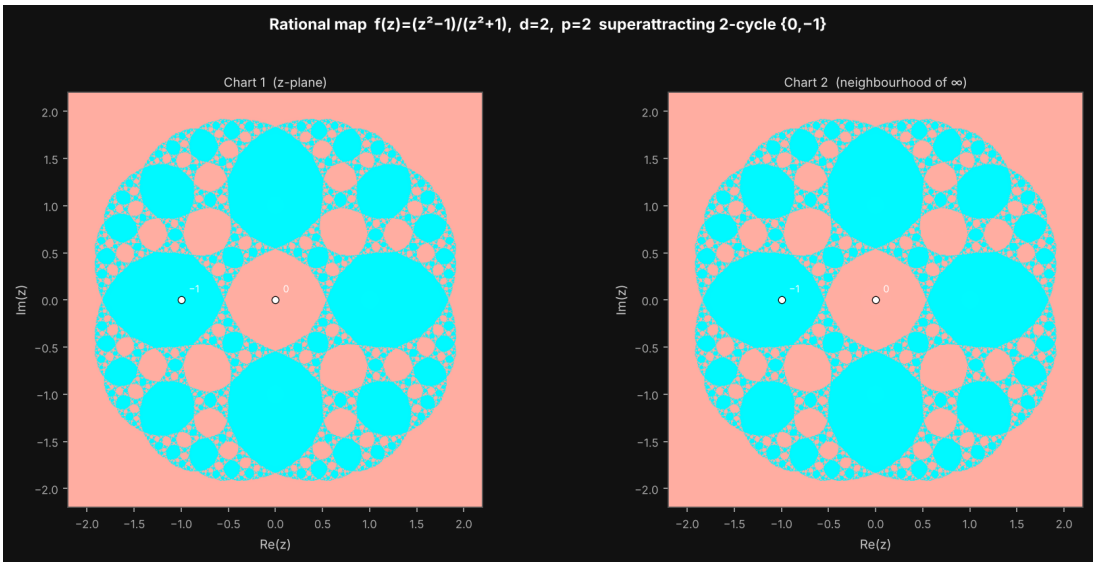


Figure 10: Basin portrait of $f(z) = (z^2 - 1)/(z^2 + 1)$, the simplest proper rational map with a superattracting 2-cycle $\{0, -1\}$. Degree $d = 2$, two critical points ($z = 0$ and $z = \infty$), both attracted to the cycle. Chart 1 (left): the two immediate basins B_0^0 (orange) and B_{-1}^0 (blue) separated by the Julia set. Chart 2 (right): the neighbourhood of ∞, which lies in B_0^0 since $\infty \to 1 \to 0$.

7.10 Summary table

Map	d	# crit. pts	P	Dominant labels
Möbius (degree 1)	1	0	0	—
$f(z) = z^2$	2	1	1	superattr. ($p = 1$)
$f(z) = z^2 - 1$	2	2	2	superattr. ($p = 2$) + superattr. ($p = 1$ at ∞)
Newton $z^3 - 1$	3	4	1	3 superattr. ($p = 1$) + 1 indet.
Airplane $z^2 + c$	2	2	3	superattr. ($p = 3$) + superattr. ($p = 1$ at ∞)
Rabbit $z^2 + c$	2	2	3	superattr. ($p = 3$) + superattr. ($p = 1$ at ∞)
Period-5 $z^2 + c$	2	2	5	superattr. ($p = 5$) + superattr. ($p = 1$ at ∞)
Halley $z^3 - 1$	4	6	1	3 superattr. ($p = 1$), 2 repelling fixed pts
$(z^2 - 1)/(z^2 + 1)$	2	2	2	1 superattr. 2-cycle (proper rational)

8 Discussion and open problems

8.1 The bound P and its sharpness

By Corollary 4.6, the bound P output by `max_attracting_period` is sharp: there is no attracting cycle of period $> P$. However, P is computed via a finite warmup (default $N_{\text{warm}} = 300$), so if the warmup is insufficient for a slowly-converging attracting cycle, P may be underestimated. A conservative use of the bound requires checking that the Lyapunov constant satisfies $c \ll 1$; near a parabolic parameter where $c \to 1$, the warmup must be increased substantially.

8.2 Beyond attracting cycles

The current algorithms detect only attracting and superattracting cycles. Parabolic cycles ($|\lambda| = 1$, $\lambda^q = 1$) also attract orbits, but at polynomial rather than exponential rate; they require a modified detection scheme based on Fatou coordinates. Siegel discs and Herman rings require a different approach (e.g., measuring the autocorrelation of the orbit to detect irrational rotation numbers).

8.3 Families and bifurcations

For a one-parameter family f_c (e.g., $z^2 + c$), the function $c \mapsto P(c)$ records the attracting period. Its level sets are the *hyperbolic components* of the Mandelbrot set $\mathcal{M}$ (Section 7.1, [6]): $P(c) = p$ on the interior of H_p, while $P(c) = 0$ on the exterior of $\mathcal{M}$ (escaping parameters) and on the boundary $\partial\mathcal{M}$ (parabolic, Siegel, Misiurewicz and other non-hyperbolic parameters).

The Mandelbrot portrait produced by `mandelbrot_portrait` (Figure 1) can therefore be read as a *parameter portrait*: each pixel colour encodes the escape speed of the critical orbit, and the near-black interior is exactly $\mathcal{M}$. The four superattracting centres studied in this article are marked explicitly on the figure; they are canonical representatives of periods 2, 3, 3, and 5.

An algorithmic study of the hyperbolic boundaries of $\mathcal{M}$ via `max_attracting_period` applied to a dense parameter grid is a natural extension of the present work, and would yield a computational counterpart to the theoretical description of $\mathcal{M}$ in [6].

8.4 Algorithmic complexity and parallelism

Both algorithms are embarrassingly parallel over grid points (Algorithm 2) and over critical points (Algorithm 1). Julia's threading model and GPU backends (via `CUDA.jl`) can accelerate the portrait computation by several orders of magnitude for high-resolution images.

8.5 Homogeneous arithmetic and exact computation

For maps with algebraic coefficients, the Wronskian roots can in principle be computed exactly (via resultant-based methods). An exact implementation would eliminate the deduplication tolerance in Algorithm 1 and give a provably correct critical point count.

9 Conclusions

We have presented a self-contained algorithmic framework for the computation of basins of attraction of p-cyclic orbits of rational maps on the Riemann sphere $\mathbf{P}^1(\mathbb{C})$, based entirely on homogeneous coordinates and the chordal metric. The key contributions are:

1. A two-chart formula for the spherical derivative $f^\#$ (Proposition 2.2) that is correct at all points of $\mathbf{P}^1(\mathbb{C})$, including $\infty = [1 : 0]$, correcting a subtle error in naive single-chart implementations (Remark 2.3).

2. Algorithm 1 (`critical_orbit_analysis`) implementing Fatou's theorem to locate all attracting cycles and compute a sharp upper bound P on the maximum attracting period.

3. Algorithm 2 (`basin_portrait_pcycle`) producing colour-coded basin portraits for any user-specified period p, with full support for the two-chart global portrait on $\mathbf{P}^1(\mathbb{C})$.

4. A suite of 94 automated tests in `RiemannBasins.jl` validating the algorithms on the quadratic family, Newton's method, and Möbius transformations.

The combination of Algorithms 1 and 2 gives a principled workflow: run Algorithm 1 first to determine P, then run Algorithm 2 only for $p \leq P$, avoiding unnecessary computation.

Acknowledgements. This work is a continuation of the program initiated in [7, 2]. Computations were performed using the Julia language [5] with the `RiemannBasins.jl` package [3]. L. J. H.-P. acknowledges support from project REGI25/61 of the Universidad de La Rioja.

References

[1] V. 'Alvarez-Aparicio. LCD.jl: Lyapunov Cycle Detector. Julia package, 2022.

[2] V. Álvarez-Aparicio, J. M. García-Calcines, L. J. Hernández-Paricio, and M. T. Rivas-Rodríguez. Algorithms for computing basins of attraction associated with a rational self-map of the Hopf fibration based on Lyapunov exponents. *Topology and its Applications*, 339:108578, 2023.

[3] V. Álvarez-Aparicio, J. M. García-Calcines, L. J. Hernández-Paricio, and M. T. Rivas-Rodríguez. RiemannBasins.jl: A Julia package for iteration of rational maps on the Riemann sphere. Preprint, ResearchGate, 2024.

[4] V. Álvarez-Aparicio, J. M. Gutiérrez-Jiménez, L. J. Hernández-Paricio, and M. T. Rivas-Rodríguez. Computing Lyapunov exponents for the study of the dynamical behaviour of Chebyshev's method on polynomials. In J. R. Torregrosa, J.-C. Cortés, and A. Vidal-Ferràndiz, editors, *Modelling for Engineering & Human Behaviour 2023*, pages 257–265. Universitat Politècnica de València, 2023. XXV Edition of the Mathematical Modelling Conference Series, València, July 11–14, 2023.

[5] Jeff Bezanson, Alan Edelman, Stefan Karpinski, and Viral B. Shah. Julia: A fresh approach to numerical computing. *SIAM Review*, 59(1):65–98, 2017.

[6] Adrien Douady and John H. Hubbard. On the dynamics of polynomial-like mappings. *Annales Scientifiques de l'École Normale Supérieure*, 18(2):287–343, 1985.

[7] I. Extremiana-Aldana, J. J. Hernández-Paricio, L. and T. Rivas-Rodríguez, M.˙ Projective models of closed surfaces and iteration of maps. *Preprint*, 2019. Universidad de La Rioja.

[8] J. I. Extremiana-Aldana, L. J. Hernández-Paricio, and M. T. Rivas-Rodríguez. Self-overlays and symmetries of Julia sets of expanding maps. *Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat.*, 2018.

[9] J. I. Extremiana-Aldana, L. J. Hernández-Paricio, and M. T. Rivas-Rodríguez. The shape of the Julia set of an expanding rational map. *Topology and its Applications*, 239:251–273, 2018. Virtual Special Issue in memory of Professor Sibe Mardešić.

[10] Mark D. Fairchild. *Color Appearance Models*. Wiley, 3rd edition, 2013.

[11] Pierre Fatou. Sur les équations fonctionnelles. *Bulletin de la Société Mathématique de France*, 47:161–271, 1919. Premier mémoire.

[12] Pierre Fatou. Sur les équations fonctionnelles. *Bulletin de la Société Mathématique de France*, 48:33–94, 1920. Deuxième mémoire.

[13] J. M. Gutiérrez-Jiménez, L. J. Hernández-Paricio, Á. A. Magreñán, and M. T. Rivas-Rodríguez. Measures of the basins of attracting n-cycles for the relaxed Newton's method. In S. Amat and S. Busquier, editors, *Advances in Iterative Methods for Nonlinear Equations*, volume 10 of *SEMA SIMAI Springer Series*, pages 211–246. Springer International Publishing, Cham, 2016.

[14] JuliaGraphics contributors. Colors.jl: Color manipulation and conversions for Julia. `https://github.com/JuliaGraphics/Colors.jl`, 2023.

[15] John Milnor. *Dynamics in One Complex Variable*, volume 160 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, third edition, 2006.

[16] Masataka Okabe and Kei Ito. Color universal design (CUD): How to make figures and presentations that are friendly to colorblind people. *J. Biochemistry and Molecular Biology*, 2008. `https://jfly.uni-koeln.de/color/`.

[17] Heinz-Otto Peitgen and Peter H. Richter. *The Beauty of Fractals: Images of Complex Dynamical Systems*. Springer-Verlag, Berlin, 1986.

[18] Michael Reed. Fatou.jl: Fatou sets in the Julia programming language. `https://github.com/chakravala/Fatou.jl`, 2017. Julia package for complex dynamics visualisation; GPU-accelerated escape-time and orbit-trap portraits.

[19] Dennis Sullivan. Quasiconformal homeomorphisms and dynamics I: Solution of the Fatou–Julia problem on wandering domains. *Annals of Mathematics*, 122(3):401–418, 1985.